

Homework 1: Thinking About Data

BEE 4850

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Note

All of the code used in this homework is accessible in both the hw01.qmd and hw01.ipynb files in my GitHub repository. I used the Jupyter notebook to write the scripts (to more easily re-run and debug) before moving finalized code to the Quarto file to generate this pdf, and so the same code is included in both files.

Problem 1 (7 points)

Problem 1.1

Conduct the above procedure to generate 50,000 simulated datasets under the skeptic's hypothesis. Plot a histogram of the results and add a dashed vertical line to show the experimental difference.

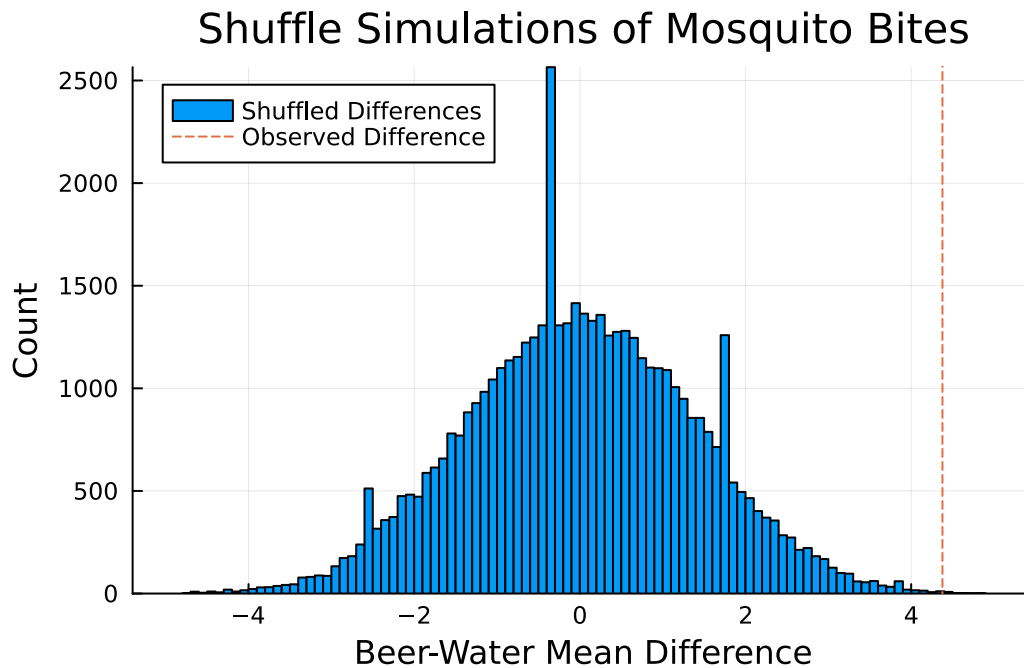


Figure 1: Histogram of Mean Differences for 50000 shuffled simulations of mosquito bite data for beer vs. water drinkers.

Problem 1.2

Draw conclusions about the plausibility of the skeptic's hypothesis that there is no difference between groups. Feel free to use any quantitative or qualitative assessments of your simulations and the observed difference.

Based on the simulation results, the skeptic's hypothesis does not appear very plausible. As would be expected from a random shuffle, the difference between the randomized "beer" and "water" category means is centered around zero, with the majority of simulations having a mean difference between -2 and 2. However, the observed difference is well over 4, and there are only a very small number of shuffles with a mean difference that large. It is unlikely that such a large difference would be drawn from this distribution of random simulations, suggesting that there is a difference in mosquito bites for beer drinkers versus water drinkers.

Problem 2 (5 points)

Problem 2.1

Load the dataset and compute the following summary statistics for each column (you can do this using built-in functions as available in the language you are using):

1. Minimum, 25th percentile, median, 75th percentile, and maximum;
2. Mean;
3. Number of missing data entries (NAs).

What are the units for the temperatures? Explain why there are so many negative values for the pollution variables.

	variable	min	q25	median	q75	max	mean	nmissing
	Symbol	Real	Float64	Float64	Float64	Real	Float64	Int64
1	Column1	1	1279.25	2557.5	3835.75	5114	2557.5	0
2	death	69	105.0	114.0	124.0	411	115.419	0
3	pm10median	-37.3761	-13.1082	-3.53906	8.30292	320.725	-0.14639	251
4	pm25median	-16.4264	-6.58841	-1.32584	5.34377	38.1504	0.243053	4387
5	o3median	-24.7794	-10.2319	-3.32586	4.46821	43.6878	-2.17938	0
6	so2median	-8.2061	-2.68935	-1.21826	0.831593	28.9034	-0.636071	27
7	time	-2556.5	-1278.25	0.0	1278.25	2556.5	0.0	0
8	tmpd	-16.0	35.0	51.0	67.0	92.0	50.1933	0

The temperatures are in units of degrees Fahrenheit, ranging from approximately -15°F to approximately 90°F, a typical range for the Midwestern United States. I was originally confused as to why there were negative values for the pollution variables, so I did a bit of online searching and found the National Ambient Air Quality Standards¹ from the U.S. Environmental Protection Agency. These standards set reference values for what “acceptable” values for air pollutants are based on public health research. Looking at the values for this standard, it seems the Chicago pollution measurements are reported in terms of their difference to this standard, with negative values below the standard and positive values above the standard.

¹<https://www.epa.gov/criteria-air-pollutants/naaqs-table>

Problem 2.2

Plot the number of deaths versus the time (convert the time to the calendar date). Describe any patterns that you see.

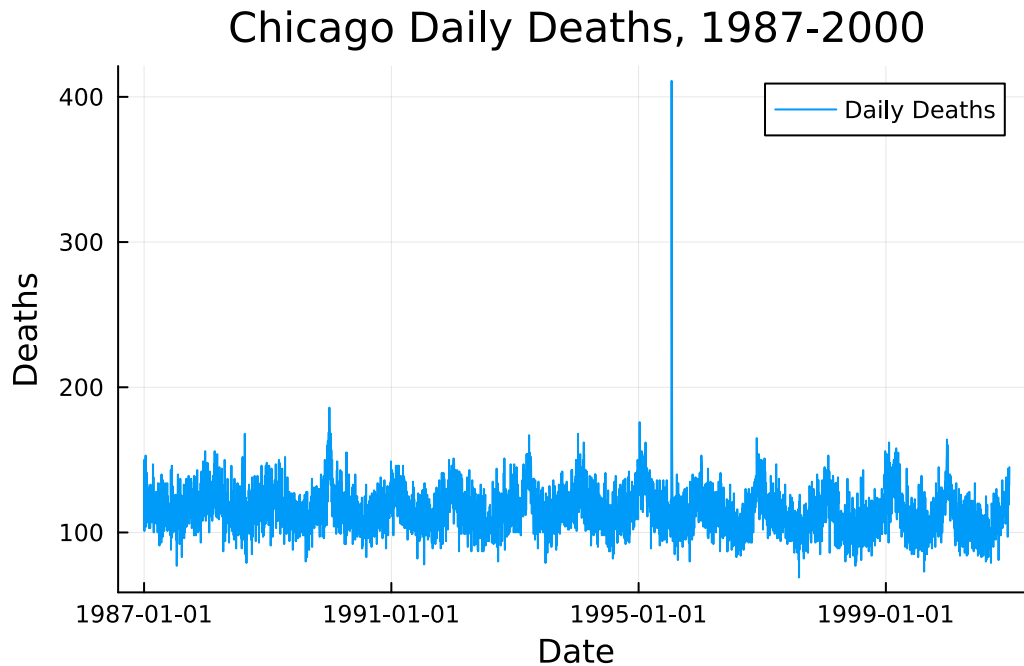


Figure 2: Plot of daily deaths in Chicago from 01-01-1987 to 12-31-2000.

There does not appear to be any significant increase or decrease in the number of deaths in Chicago throughout the 13-year period. Typically, the number of daily deaths appears to peak in the winter, decrease through the spring, reach a minimum in the summer, and then increases through the fall. Apart from a small number days, including one day with a daily total of over 400 deaths, the daily counts typically range in the low 100s, which matches what we see from the summary statistics with a q25 of 105 and a q75 of 124.

Problem 2.3

Plot the number of deaths versus the temperature. Describe any patterns that you see.

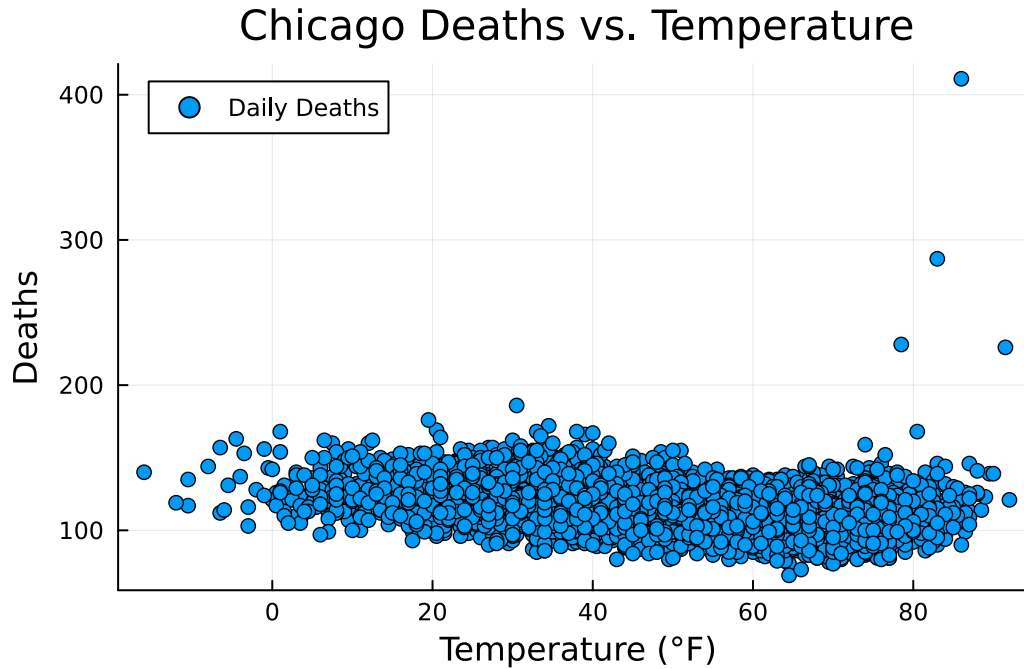


Figure 3: Scatterplot of daily deaths and temperature in Chicago from 01-01-1987 to 12-31-2000.

There is a slightly negative correlation between temperature and number of deaths, with almost no days having less than 100 deaths when the temperature is below 20°F but many such days when the temperature is above 60°F. However, the difference between daily deaths when the temperature is below freezing versus very warm does not appear to be large. However, the high outliers in terms of number of deaths all occur when the average daily temperature was extremely high, which suggested to me that this was associated with some kind of heat wave. Matching with the dates in Figure 2, I was able to find information about the July 1995 Chicago Heat Wave, which lead to over 500 deaths in the city of Chicago and caused heat index values of above 115°F.²

²https://www.weather.gov/lot/1995_HeatWave

Problem 2.4

What do you think has a clearer influence on the number of deaths: date or temperature?

I think date has a clearer influence on the number of deaths. While there does appear to be a slight negative correlation in the temperature data (Figure 3), the magnitude of the influence is difficult to discern. However, in Figure 2, the oscillating pattern of increasing and decreasing deaths with season is easily evident, and so I believe the date can be more easily used to understand the number of deaths.

Problem 3 (3 points)

Problem 3.1

Load the data file and produce the following summary statistics for the day (not the date) of full flowering (you can do this using built-in functions as available in the language you are using):

1. Minimum, 25th percentile, median, 75th percentile, and maximum;
2. Mean;
3. Number of missing data entries (NAs).

Why would it not make sense to calculate these summary statistics for the date column?

Problem 3.2

Plot the day of flowering against the year. Add horizontal lines showing the mean date of flowering as well as the 25th and 75th percentiles. Do you see any trends or patterns in the data?

Problem 4 (5 points)

A scientific funding agency receives 200 proposals in response to a call. The panel is asked to evaluate each proposal on two criteria: scientific rigor and potential impact (or “newsworthiness”). After standardizing each of these scores to independently follow a standard normal distribution ($\text{Normal}(0, 1)$), the two standardized scores are summed to get the total score for the proposal. Based on these total scores, the top 10% of the proposals are selected for funding.

A researcher who is studying the relationship between rigor and impact has used data on the funded proposals to claim high-impact proposals are necessarily less rigorous, and indeed found a statistically significant negative correlation between the rigor and impact scores for the funded proposals. You are more skeptical and believe this effect is an artifact from the selection process, which would make the claim a bit ironic.

Problem 4.1

Create a generative model for the grant-selection procedure under the null assumption of no correlation between rigor and impact. You can sample rigor and impact scores directly from $\text{Normal}(0, 1)$. Generate 1,000 datasets using this model and flag the proposals which would be funded in each. Plot a histogram of the correlations between rigor and data for (a) the funded proposals and (b) the overall pool. What do you observe?

Problem 4.2

Explain why, when conditioning on selection for funding, there might be a negative correlation between rigor and impact even if there none generally exists.

! Selection-Distortion Effects

These **selection-distortion effects** are pretty common in observational data, going back to [work by Dawes in the 1970s on the lack of predictive ability of admission variables on student success](#) and including [studies claiming to identify a causal fingerprint of genes on outcomes](#).

This is, of course, just one example of how not thinking carefully about data-generating processes can fundamentally contaminate statistical analyses, emphasizing that **data do not have meaning absent a model for how they were generated**.

Problem 5 (5 points)

You are trying to detect how prevalent cheating was on an exam. You are skeptical of the efficacy of just asking the students if they cheated. You are also concerned about privacy — your goal is not to punish individual students, but to see if there are systemic problems that need to be addressed. Someone proposes the following interview procedure, which the class agrees to participate in:

Each student flips a fair coin, with the results hidden from the interviewer. The student answers honestly if the coin comes up heads. Otherwise, if the coin comes up tails, the student flips the coin again, and answers “I did cheat” if heads, and “I did not cheat”, if tails.

You have a hypothesis that cheating was not prevalent, and the proportion of cheaters was no more than 5% of the class; in other words, we expect 5 “true” cheaters out of a class of 100 students. Our TA is more jaded and thinks that cheating was more rampant, and that 30% of the class cheated. The proposed interview procedure is noisy: the interviewer does not know if an admission means that the student cheated, or the result of a heads. However, it gives us a data-generating process that we can model and analyze for consistency with our hypothesis and that of the TA.

Problem 5.1

Derive and code a simulation model for the above interview procedure given a “true” probability of cheating p . Simulate your model (for a class of 100 students) 50,000 times under your hypothesis of 5% cheating, the TA’s hypothesis of 30% cheating, and plot the resulting datasets.

Problem 5.2

How many “Yes, I cheated” answers do you think you’d need to see to feel confident concluding that the TA was right and why? You don’t need to do a formal analysis, but can reason about this based on your plots.

Aono, Y., & Omoto, Y. (1993). Variation in the March mean temperature deduced from cherry blossom in Kyoto since the 14th century. *Journal of Agricultural Meteorology*, 48(5), 635–638. <https://doi.org/10.2480/agrmet.48.635>

Aono, Y., & Omoto, Y. (1994). Estimation of temperature at Kyoto since the 11th century. Using flowering data of cherry trees in old documents. *Journal of Agricultural Meteorology*, 49(4), 263–272. <https://doi.org/10.2480/agrmet.49.263>

Aono, Y., & Kazui, K. (2008). Phenological data series of cherry tree flowering in Kyoto, Japan, and its application to reconstruction of springtime temperatures since the 9th century. *International Journal of Climatology: A Journal of the Royal Meteorological Society*, 28(7), 905–914. <https://doi.org/10.1002/joc.1594>